

Final Report – Heart Disease Analysis

1. INTRODUCTION

1.1 Project Overview

Heart Disease Analysis is a healthcare data analytics project developed using Tableau. The project analyses patient health records to identify patterns and risk factors associated with heart disease. Interactive visualizations, dashboards, and stories help users explore relationships between demographic, lifestyle, and medical factors such as age, gender, diabetes, smoking, alcohol consumption, BMI, race, physical activity, and general health. The project supports healthcare professionals in making informed decisions through data-driven insights.

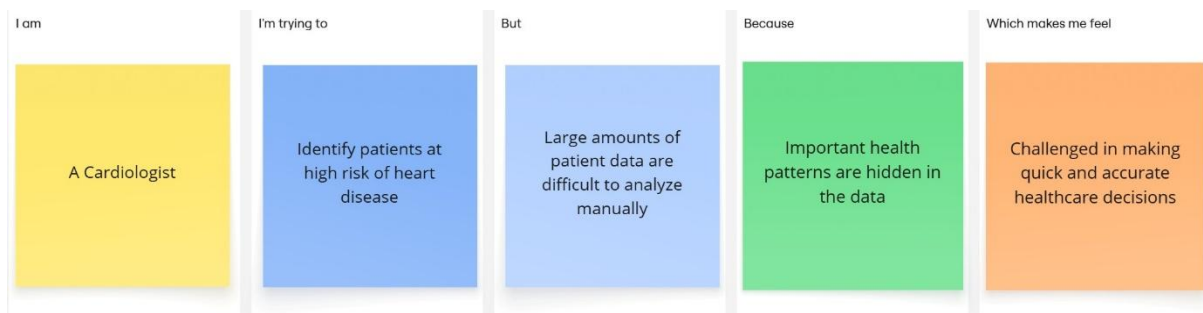
1.2 Purpose

The purpose of this project is to transform raw healthcare data into meaningful visual insights using Tableau. The dashboard enables users to identify high-risk patient groups, analyse disease trends, and understand the impact of various health factors on heart disease.

2. IDEATION PHASE

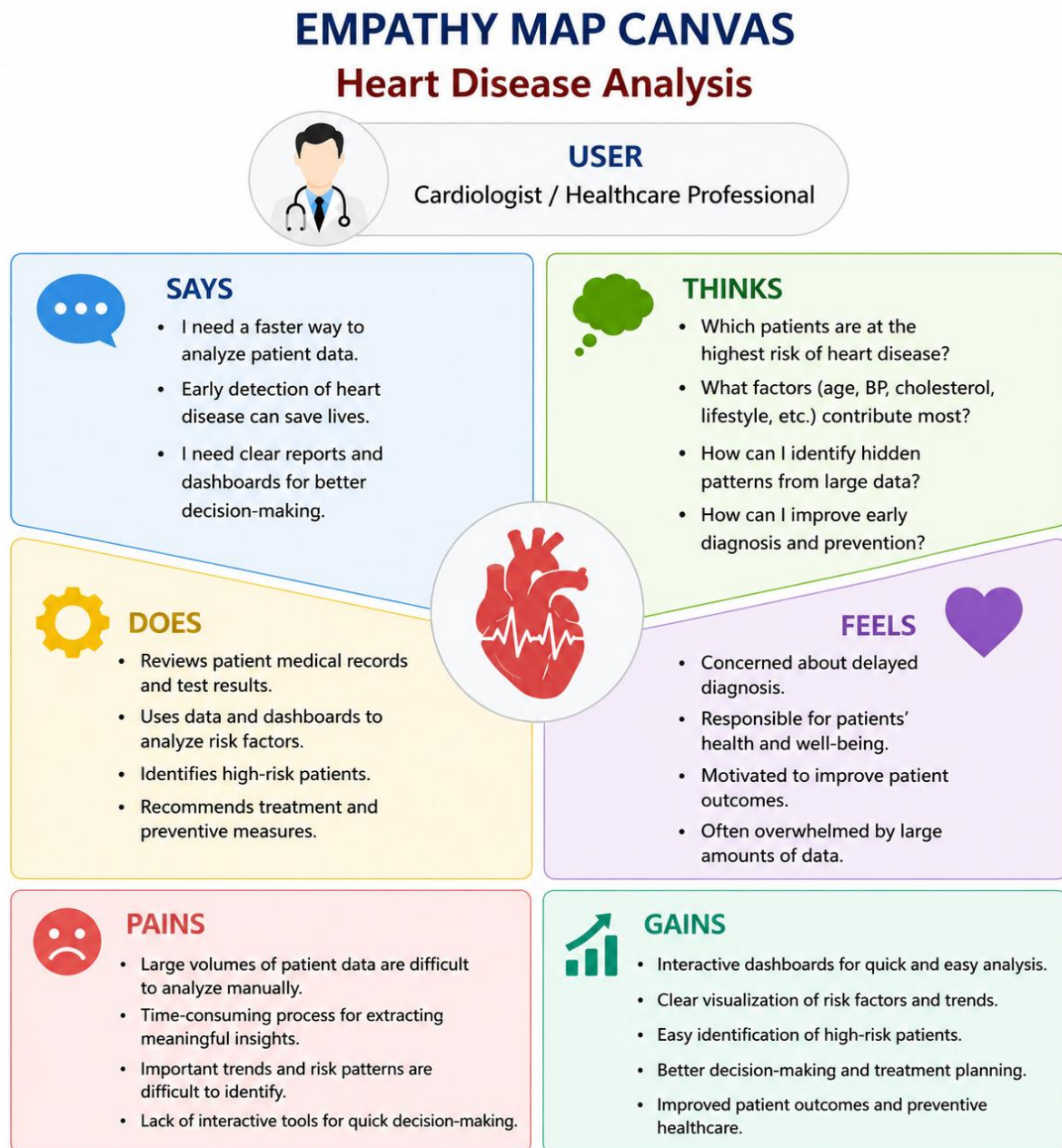
2.1 Problem Statement

Heart disease is one of the leading causes of death worldwide. Healthcare professionals often find it difficult to analyse large volumes of patient data and identify key risk factors through manual methods. This project provides an interactive Tableau dashboard to simplify data analysis and support better healthcare decision-making.



2.2 Empathy Map Canvas

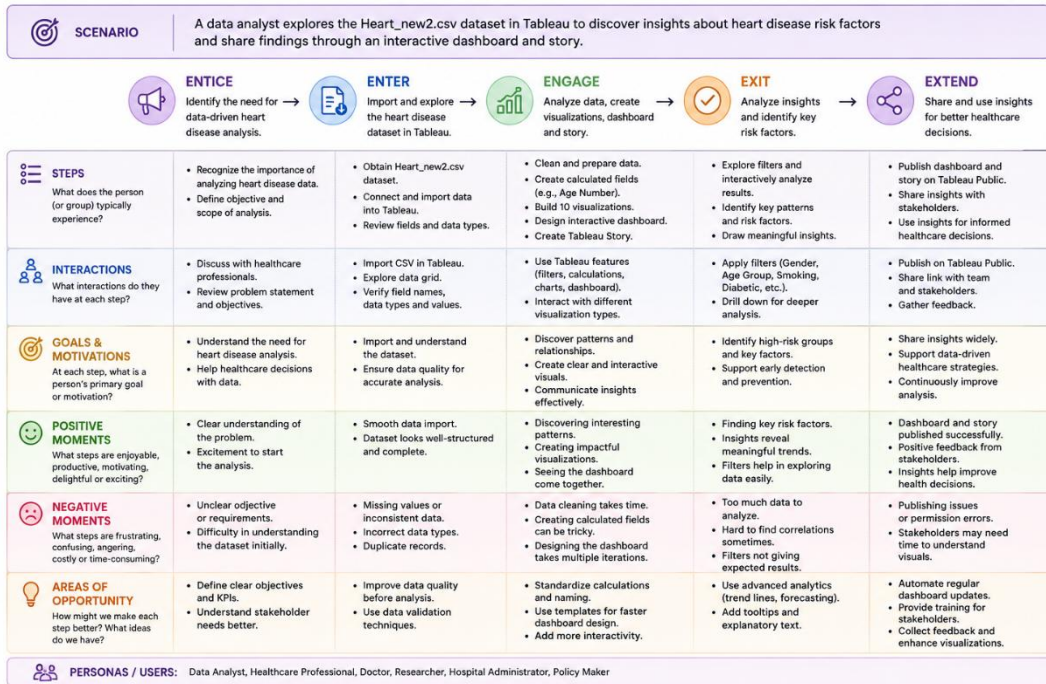
The Empathy Map was created to understand the needs, challenges, and expectations of healthcare professionals and other stakeholders involved in heart disease analysis.



2.3 Brainstorming

Brainstorming sessions generated ideas related to healthcare analytics, data visualization, and interactive dashboards. The best solution—Heart Disease Analysis using Tableau—was selected through idea prioritization.

CUSTOMER JOURNEY MAP HEART DISEASE ANALYSIS USING TABLEAU



3.2 Solution Requirement

The solution requires a healthcare dataset in CSV format, Tableau for visualization, and an interactive dashboard to present meaningful insights.

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import the Heart Disease CSV dataset into Tableau Validate and verify the dataset before analysis
FR-2	Data Preparation	Clean and preprocess the data Handle missing values and assign correct data types
FR-3	Data Visualization	Create interactive charts and graphs Generate visualizations for heart disease analysis
FR-4	Dashboard & Story	Design an interactive dashboard Create Tableau Story with insights and conclusions
FR-5	Data Filtering & Analysis	Apply filters based on Age, Gender, Race, Diabetes, Smoking, and Heart Disease Enable users to analyze specific patient groups using interactive filters
FR-6	Report Generation	Publish the Tableau workbook and dashboard Generate project documentation and share insights through GitHub and Tableau Public

Non-functional Requirements:

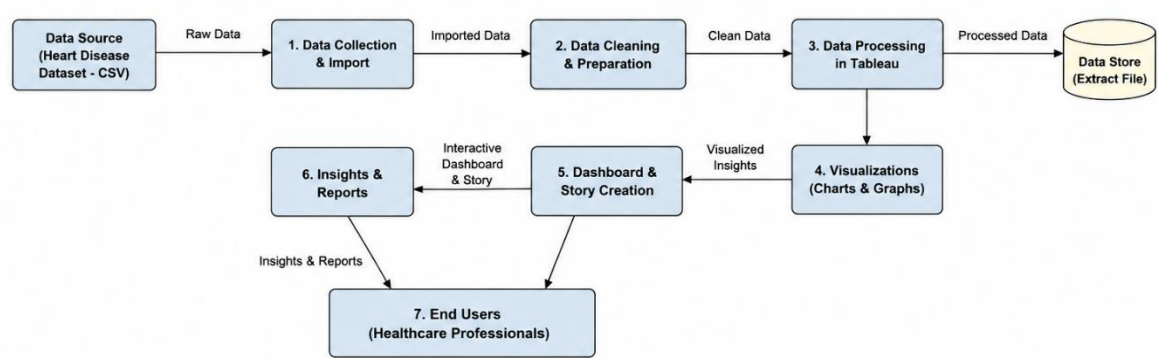
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to understand and navigate.
NFR-2	Security	The dataset should be securely stored and used only for analysis.
NFR-3	Reliability	The dashboard should display accurate and consistent results.
NFR-4	Performance	Visualizations should load quickly and respond efficiently.
NFR-5	Availability	The Tableau dashboard should be available whenever required.
NFR-6	Scalability	The solution should support larger datasets in the future.

3.3 Data Flow Diagram

The Data Flow Diagram illustrates the flow of data from the CSV dataset through preprocessing, visualization, dashboard creation, and final insights.

Data Flow Diagram (Level 0) – Heart Disease Analysis



3.4 Technology Stack

Software Used

- Tableau Public
- Tableau Desktop

- Microsoft Excel
- GitHub

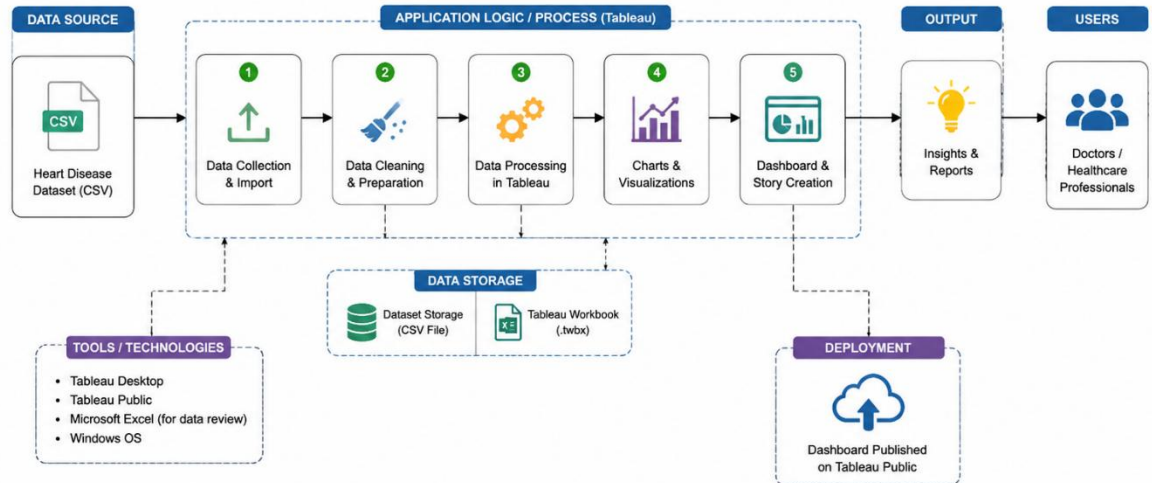
Dataset

- Heart_new2.csv

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Technical Architecture – Heart Disease Analysis



- All processes (application logic / technology blocks) are represented.
- Infrastructural demarcation (Local / Cloud) is indicated.
- External interfaces – Tableau Public (for publishing dashboards).
- Data storage components – CSV dataset and Tableau workbook.
- (No machine learning model required for this project.)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution effectively addresses the challenges of manual healthcare data analysis by providing interactive visualizations and dashboards that support faster and more accurate decision-making.

Problem – Solution Fit Template:

Define & Jobs (Left Side)	1. CUSTOMER SEGMENTS (CS) Who is your customer? For whom are you solving the problem? - Healthcare Professionals - Cardiologists - Doctors - Hospital Administrators - Public Health Officials - Medical Researchers	CS	6. CUSTOMER CONSTRAINTS (CC) What are the constraints (time, resources, tools, skills, cost) that prevent your customers from finding a better solution? - Limited time for data analysis - Large and complex datasets - Lack of interactive visualization tools - Difficulty identifying hidden trends - Need for accurate and reliable insights	CC	5. AVAILABLE SOLUTIONS (AS) What solutions are available for the customer when they face this problem? - Manual spreadsheet analysis - Static reports - Basic statistical tools - Traditional hospital records - General health surveys	AS	Explore AS & Constraints		
	2. JOBS-TO-BE-DONE / PROBLEMS (JBP) What job(s) should be done / problem(s) that your customer is trying to solve? - Analyze large heart disease datasets - Identify high-risk patients - Understand lifestyle risk factors - Generate meaningful healthcare insights - Support preventive healthcare decisions	JBP	9. PROBLEM ROOT CAUSE (RC) What is the true cause of the customer's problem? What is the underlying factor that leads to this issue? - Growing lifestyle-related health issues - Smoking and alcohol consumption - Diabetes and obesity - Physical inactivity - Poor general health awareness	RC	7. BEHAVIOUR (BE) How does your customer try to address the problem right now? List their key actions, habits or workflow. - Collect patient health records - Analyze health indicators - Compare different patient groups - Monitor disease trends - Generate healthcare reports	BE		Focus on Behaviour	
Focus on Jobs & Problems	3. TRIGGERS (TR) What triggers (internal or external) make your customer aware of the problem? - Increasing heart disease cases - Need for faster diagnosis - Growing healthcare data - Demand for visual data analysis - Awareness of preventive healthcare	TR	4. EMOTIONS BEFORE / AFTER (EM) How does the customer feel before facing the problem? And how do they feel after solving it? Before: - Difficult to interpret large datasets - Time-consuming manual analysis - Uncertainty in identifying risk factors After: - Quick understanding of patient data - Easy identification of high-risk groups - Increased confidence in decision-making	EM	10. YOUR SOLUTION (SL) What is your solution to the customer's problem? How does it solve their job? Develop an interactive Heart Disease Analysis Dashboard using Tableau that imports and analyzes patient data from a CSV dataset. The dashboard provides visual insights into age, gender, diabetes, smoking, alcohol consumption, BMI, race, physical activity, and general health. Interactive charts, dashboards, and stories help healthcare professionals identify high-risk groups and support informed medical decision-making.	SL	8. CHANNELS & MESSAGING (CH) Through which channels will you reach your customer? What is the key message that will attract and engage them? Channels: - Tableau Dashboard - Tableau Story - Tableau Public - GitHub Repository Messaging: - Interactive visualization for better healthcare insights. - Data-driven decision support. - Early identification of heart disease risk.		CH
	Identify Triggers								

4.2 Proposed Solution

The proposed solution imports the Heart Disease dataset into Tableau, performs preprocessing, creates interactive visualizations, and presents healthcare insights through dashboards and stories.

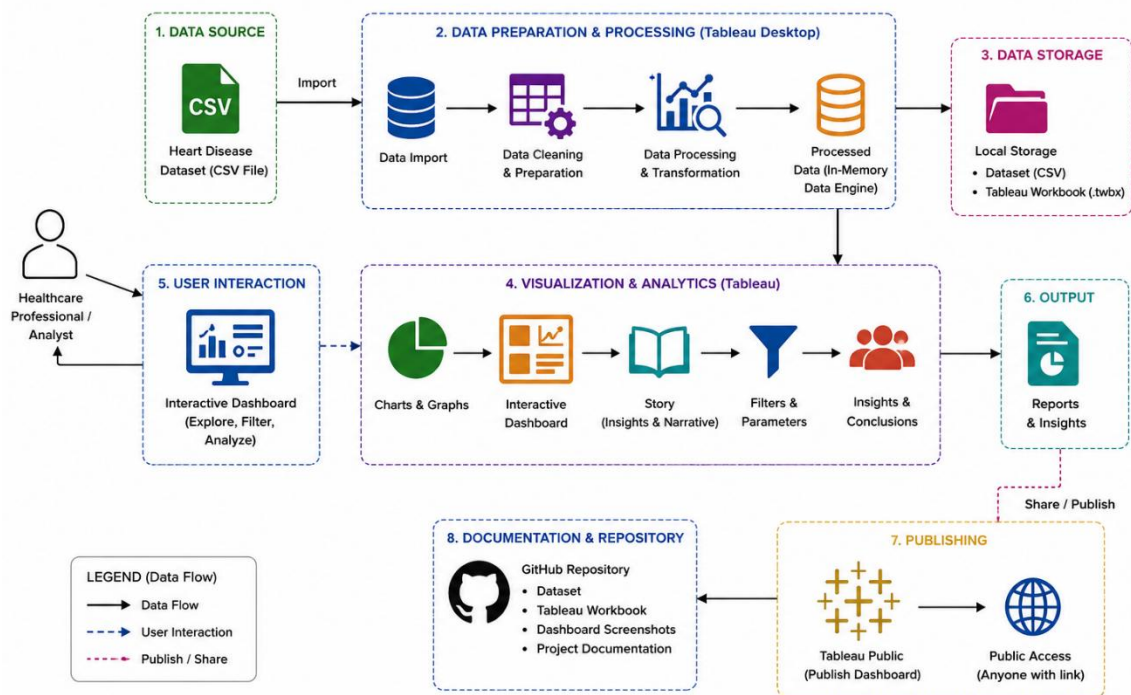
Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Heart disease is one of the leading causes of death worldwide. Healthcare professionals often find it difficult to analyze large volumes of patient data and identify key risk factors such as age, gender, diabetes, smoking, alcohol consumption, BMI, race, and physical activity. Manual analysis is time-consuming and may overlook important patterns needed for early diagnosis and preventive care.
2.	Idea / Solution description	Develop an interactive Heart Disease Analysis Dashboard using Tableau. The dashboard imports patient data from a CSV dataset, performs data visualization, and presents interactive charts, dashboards, and stories that help healthcare professionals identify high-risk groups, analyze disease trends, and make informed decisions.
3.	Novelty / Uniqueness	The solution combines multiple health indicators into a single interactive dashboard. Users can apply filters, compare patient groups, and explore insights through visualizations and stories, making healthcare analysis faster, easier, and more effective.
4.	Social Impact / Customer Satisfaction	The project promotes awareness of heart disease risk factors and supports early diagnosis. Healthcare professionals can make quicker and more informed decisions, while patients benefit from better preventive care and improved health outcomes.
5.	Business Model (Revenue Model)	The dashboard can be offered to hospitals, healthcare organizations, research institutes, and government health departments as a data analytics solution. Revenue can be generated through software licensing, dashboard customization, training services, annual maintenance, and consulting.
6.	Scalability of the Solution	The solution is scalable and can easily accommodate larger datasets, additional health parameters, multiple hospitals, or integration

4.3 Solution Architecture

The solution architecture consists of data collection, preprocessing, visualization, dashboard creation, and story generation, resulting in actionable healthcare insights.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

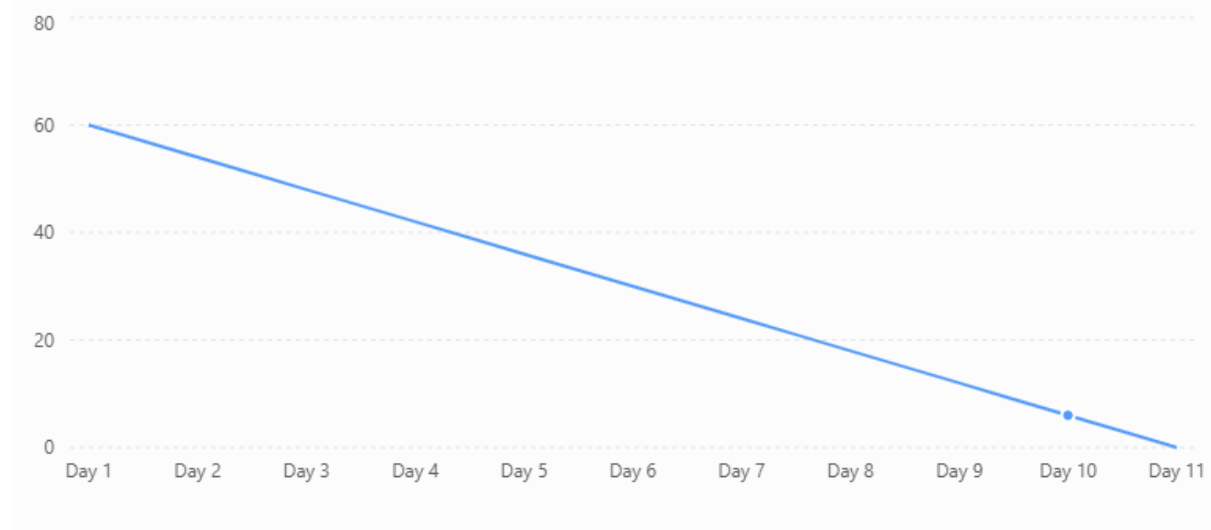
The project was executed using sprint-based planning. Tasks included dataset preparation, visualization development, dashboard design, story creation, documentation, testing, and final deployment.

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	Import Heart Disease CSV dataset into Tableau	2	High	Nikshitha Ramesh
Sprint-1	Data Preparation	USN-2	Clean and preprocess the dataset	2	High	Abdul Khayyum
Sprint-1	Data Preparation	USN-3	Verify data types and remove inconsistencies	1	Medium	Harsha
Sprint-2	Data Visualization	USN-4	Create Gender vs Heart Disease visualization	2	High	Siddesh
Sprint-2	Data Visualization	USN-5	Create remaining Tableau visualizations	3	High	Nikshitha Ramesh
Sprint-2	Dashboard	USN-6	Design an interactive Tableau Dashboard	3	High	Nikshitha Ramesh
Sprint-3	Story	USN-7	Create Tableau Story with insights	2	High	Nikshitha Ramesh
Sprint-3	Publishing	USN-8	Publish dashboard on Tableau Public	1	Medium	Siddesh
Sprint-3	Documentation	USN-9	Prepare GitHub repository and project documentation	3	High	Nikshitha Ramesh
Sprint-4	Project Demonstration	USN-10	Record demo video and complete final submission	2	High	Abdul Khayyum

Heart Disease Analysis - Sprint Burndown Chart

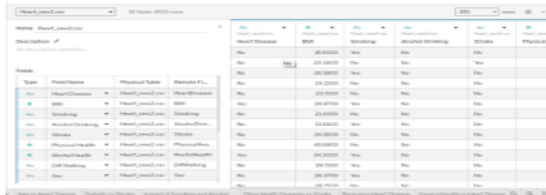
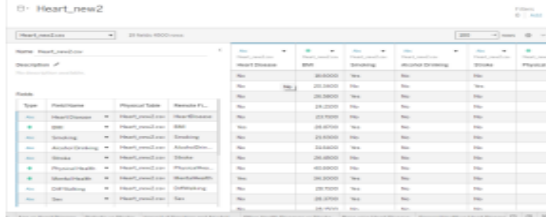
Remaining story points over the project duration.



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing verified successful data import, preprocessing, utilization of filters, calculated fields, dashboard design, and story creation. The dashboard efficiently analysed 4,500 patient records and produced interactive visualizations.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Dataset: Heart_new2.csv Total Records: 4,500 Total Columns: 19 Heart_new2 
2.	Data Preprocessing	The Heart_new2.csv dataset was imported into Tableau. Field names and data types were verified, and the data was checked for consistency before creating visualizations, dashboards, and stories. Heart_new2 

3.	Utilization of Filters	Filters used: Gender, Age Category, Heart Disease, Diabetic, Stroke, Race, Smoking, Physical Activity, General Health. Example: People with Stroke, Heart Disease and Diabetic  Heart Disease: ☒ No ☒ Yes Stroke: ☐ (All) ☐ No ☒ Yes Previous
4.	Calculation fields Used	Number of Calculated Fields: 1(Age Number)  CASE: [Age: Category] WHEN "18-24" THEN 21 WHEN "25-29" THEN 27 WHEN "30-34" THEN 32 WHEN "35-39" THEN 37 WHEN "40-44" THEN 42 WHEN "45-49" THEN 47 WHEN "50-54" THEN 52 WHEN "55-59" THEN 57 WHEN "60-64" THEN 62 WHEN "65-69" THEN 67 WHEN "70-74" THEN 72 The calculation is valid. 5 Dependencies Apply OK
5.	Dashboard design	Number of Dashboards: 2 Number of Visualizations: 10 
6	Story Design	Number of Story Points: 10 Visualizations Included: 10 

7. RESULTS

7.1 Output Screenshots

- Dashboard Screenshot:



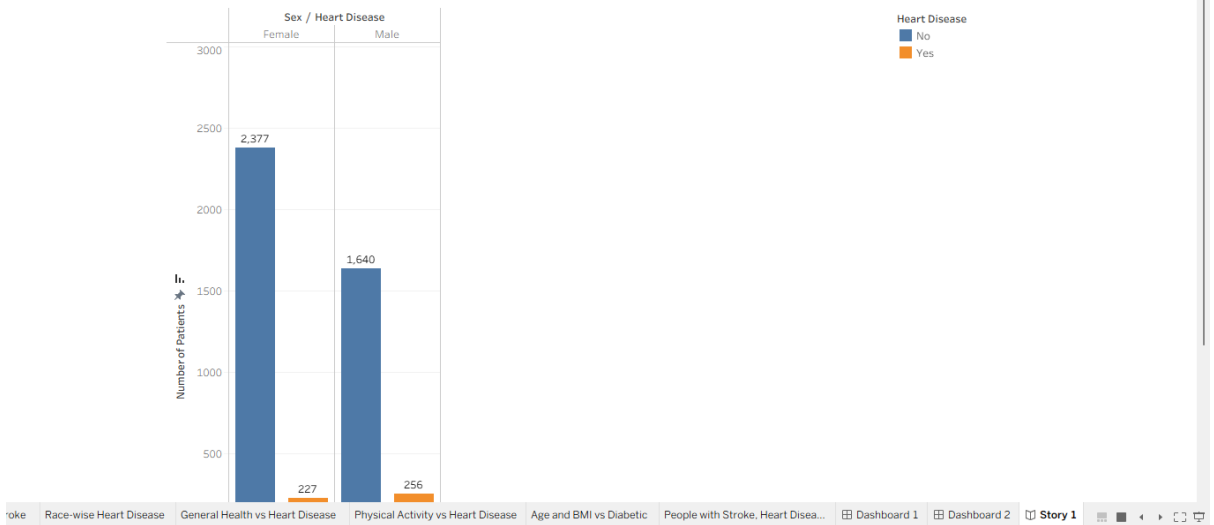
- Story Screenshot:

Heart Disease Analysis

This visualization compares heart disease distribution among males and females. It helps identify gender-based differences in heart disease occurrence.

This analysis shows how heart disease prevalence changes across different age groups.

This analysis examines the relationship between diabetes and stroke. It helps identify how stroke vary among different diabetic conditions.

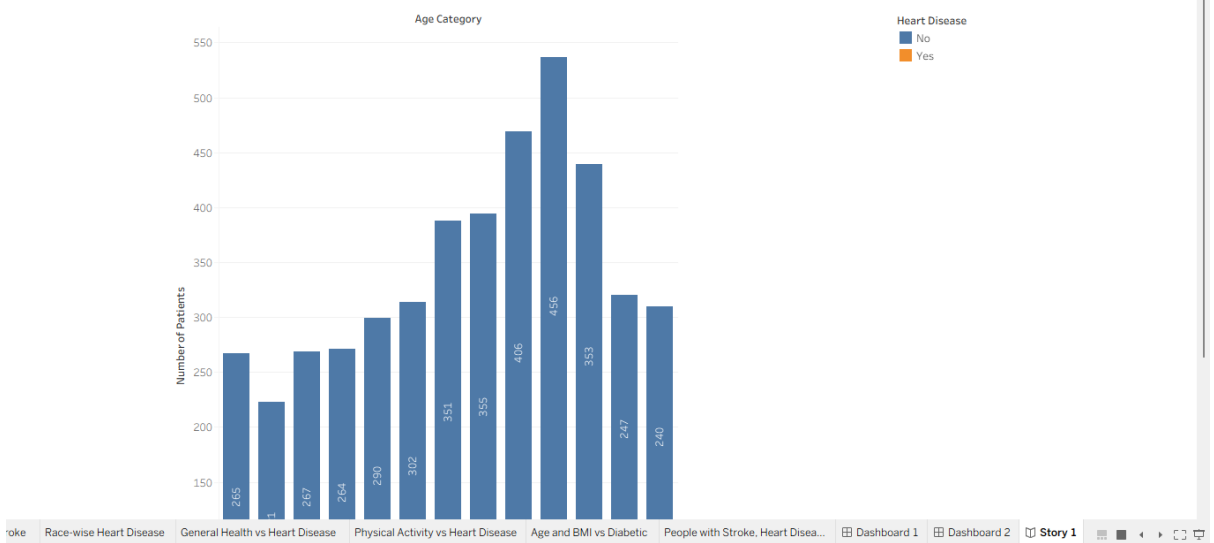


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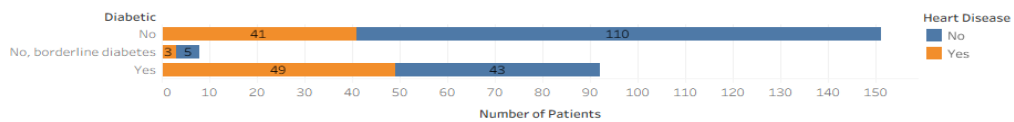


Heart Disease Analysis

This visualization compares physically active and inactive patients to understand the impact of physical activity on heart disease.

This visualization examines the relationship between age, body mass index (BMI), and diabetic status, helping identify trends among diabetic patients.

This visualization summarizes patients who have experienced stroke while also having heart disease and diabetes, highlighting the combined effect of multiple health conditions.



8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboard for healthcare analysis.
- Easy identification of heart disease risk factors.
- Supports data-driven decision-making.
- User-friendly visualization of large datasets.
- Efficient analysis using filters and stories.

Disadvantages

- Depends on the quality of the dataset.
- Limited to the available patient records.
- Does not predict future heart disease risk.
- Requires Tableau software to edit the project.

9. CONCLUSION

The Heart Disease Analysis project successfully transformed healthcare data into meaningful visual insights using Tableau. Interactive dashboards and stories helped identify relationships between demographic, lifestyle, and medical factors influencing heart disease. The project demonstrates how data visualization can improve healthcare analysis and support informed medical decision-making.

10. FUTURE SCOPE

- Integrate real-time healthcare datasets.
- Apply machine learning for heart disease prediction.
- Deploy dashboards on cloud platforms.
- Integrate with hospital management systems.
- Develop mobile-friendly healthcare dashboards.

11. APPENDIX

Dataset Link

https://github.com/Nikkiramesh424/skillwallet-project/blob/main/Heart_Disease_Analysis/Project%20Development%20Phase/Heart_new2.csv

GitHub Repository

<https://github.com/Nikkiramesh424/skillwallet-project>

Tableau Dashboard

https://public.tableau.com/views/Heart_Disease_Analysis_Dashboard_17829211184720/Dashboard2

Tableau Story

<https://public.tableau.com/shared/G8XNQNT2>

Demo Video

https://drive.google.com/file/d/1HMVOyqc5nra1MlpRsD_gUrHzHxd-tJ4S/view?usp=drivesdk